

SAI PRANEETH BHATTU

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EDUCATION

University of Michigan-Dearborn

Master's in Computer and Information Science

Jan 2026 - Dec 2027

Dearborn, MI

CMR Engineering College

Bachelor of Technology in Computer Science and Engineering

Nov 2021 - July 2025

Hyderabad, India

SKILLS

Cloud Platforms: AWS (EC2, VPC, RDS, S3, IAM, Auto Scaling, ALB, CloudWatch)

DevOps and CI/CD: Microservices Architecture, Docker, Kubernetes, Terraform, Jenkins, Ansible, GitHub Actions, SonarQube

Monitoring and Observability: Prometheus, Grafana

Networking: CIDR/Subnetting, NAT Gateway, VPC Peering, Network ACLs

Operating Systems and Scripting: Linux/Unix, Shell Scripting, Windows, SSH Key Management, RPM/YUM/APT Package Managers

Programming Languages: Python, JavaScript

Web Technologies: HTML5, CSS3, React, Node.js, Vanilla CSS, MySQL

Machine Learning: TensorFlow, Keras, LSTM, Time-Series Modeling, Scikit-learn

EXPERIENCE

AWS Academy - Virtual Internship

Cloud and Machine Learning Intern

May 2023 - July 2023

Remote

- Completed a 10-week program deploying end-to-end ML models on AWS cloud infrastructure, improving model deployment efficiency through automated pipelines.
- Implemented **automated model deployment pipelines** using AWS services (EC2, S3, IAM), reducing manual deployment steps by configuring scalable, reliable cloud workflows.
- Configured **Prometheus** for resource monitoring across cloud instances and built **Grafana** dashboards to visualize infrastructure metrics and alert on anomalies.
- Earned **AWS AI and Cloud Computing Certification** upon program completion via Edu Skills Foundation.

Code Clause - Web Development Internship

Frontend Developer Intern

Oct 2023 - Nov 2023

Remote

- Built and deployed **3 responsive frontend modules** using HTML5, CSS3, and JavaScript, improving cross-device compatibility across mobile and desktop viewports.
- Optimized UI performance in live production projects by refactoring CSS with modern frameworks, reducing load time and improving user experience scores.

PROJECTS

3-Tier Web Architecture | AWS, EC2, VPC, RDS, ALB, Auto Scaling, Nginx, Node.js | [GitHub](#)

- Designed and deployed a production-grade **3-tier architecture** on AWS spanning Web (Nginx), Application (Node.js), and Database (RDS/MySQL) tiers across **6 subnets in 2 Availability Zones** for high availability.
- Configured **Application Load Balancer** with Auto Scaling Groups to handle variable traffic; enforced least-privilege access via **IAM roles, Security Groups, and Network ACLs** between tiers.
- Implemented **multi-AZ deployment** with automated RDS backups and EC2 snapshot policies to achieve fault tolerance and zero-data-loss recovery objectives.

CI/CD Multi-Server Pipeline | Jenkins, Ansible, SonarQube, Tomcat, GitHub | [GitHub](#)

- Built an end-to-end **CI/CD pipeline** integrating GitHub, Jenkins, SonarQube, and Ansible to automate build, code quality analysis, and deployment across multiple target Linux servers.
- Authored **Ansible playbooks** (YAML) to automate Apache Tomcat 10.x installation, configuration deployment, and service management across remote Linux hosts, eliminating manual server setup.
- Integrated **SonarQube** for static code analysis into the Jenkins pipeline, enforcing code quality gates before artifact promotion to staging servers.

Stock Market Forecasting (LSTM) | *Python, TensorFlow, Keras, Streamlit* | [Live App](#)

- Developed an **LSTM-based time-series forecasting system** in Python using TensorFlow/Keras, integrating technical indicators (RSI, MACD, Bollinger Bands) to predict stock prices with measurable accuracy gains over baseline models.
- Deployed the model as an interactive **Streamlit web application**, enabling real-time stock selection, forecast visualization, and pattern analysis for end users.
- **Research published** at the 2nd International Conference on Advances in Computational Intelligence Technologies (ICACIT-2025); accepted in a **Scopus-indexed proceeding**.

RESEARCH AND PUBLICATION

Stock Market Prediction Using Multi-Filtered LSTM Approach

- Presented at the **2nd International Conference on Advances in Computational Intelligence Technologies (ICACIT-2025)**, Technocrats Institute of Technology, Bhopal, India - June 2025.
- Paper accepted for publication in a **Scopus-indexed proceeding**; proposed a multi-filtered LSTM model using technical and sentiment signals to improve forecast precision and new signal exposures over baseline models.

CERTIFICATIONS AND ACHIEVEMENTS

- **AWS AI and Cloud Computing Certification** - Edu Skills Foundation (AWS Academy)
- **Machine Learning and Python Certification** - Spoken Tutorial, IIT Bombay
- **Full Stack Web Development** - Udemy (Dr. Angela Yu)
- **Python Bootcamp** - Udemy (Abdul Bari)
- **TCS ion Career Edge Professional** - Tata Consultancy Services
- **Runner-Up**, Smart India Hackathon (SIH) Internal Round 2023